



## Exercise-1

Marked Questions can be used as Revision Questions.

### PART - I : SUBJECTIVE QUESTIONS

#### Section (A) : Equation of SHM

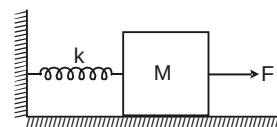
- A-1.** The equation of a particle executing SHM is  $x = (5\text{m})\sin\left[(\pi\text{s}^{-1})t + \frac{\pi}{6}\right]$ . Write down the amplitude, initial phase constant, time period and maximum speed.
- A-2.** A particle having mass 10 g oscillates according to the equation  $x = (2.0\text{ cm}) \sin [(100\text{ s}^{-1})t + \pi/6]$ . Find  
(a) the amplitude, the time period and the force constant  
(b) the position, the velocity and the acceleration at  $t = 0$ .
- A-3.** A simple harmonic motion has an amplitude  $A$  and time period  $T$ . Find the time required by it to travel directly from  
(a)  $x = 0$  to  $x = A/2$       (b)  $x = 0$  to  $x = \frac{A}{\sqrt{2}}$       (c)  $x = A$  to  $x = A/2$       (d)  $x = -\frac{A}{\sqrt{2}}$  to  $x = \frac{A}{\sqrt{2}}$   
(e)  $x = \frac{A}{\sqrt{2}}$  to  $x = A$ .
- A-4.** A particle is executing SHM with amplitude  $A$  and has maximum velocity  $v_0$ . Find its speed when it is located at distance of  $A/2$  from mean position.
- A-5.** A particle executes simple harmonic motion with an amplitude of 10 cm and time period 6 s. At  $t = 0$  it is at position  $x = 5\text{ cm}$  from mean position and going towards positive  $x$ -direction. Write the equation for the displacement  $x$  at time  $t$ . Find the magnitude of the acceleration of the particle at  $t = 4\text{ s}$ .
- A-6.** A particle is executing SHM. Find the positions of the particle where its speed is 8 cm/s, If maximum magnitudes of its velocity and acceleration are 10 cm/s and 50 cm/s<sup>2</sup> respectively.

#### Section (B) : Energy

- B-1.** A particle performing SHM with amplitude 10 cm. At What distance from mean position the kinetic energy of the particle is thrice of its potential energy?
- B-2.** An object of mass 0.2 kg executes simple harmonic oscillations along the  $x$ -axis with a frequency of  $(25/\pi)\text{ Hz}$ . At the position  $x = 0.04\text{ m}$ , the object has kinetic energy of 0.5 J and potential energy 0.4 J. Find the amplitude of oscillations [ 1994 ; 2M ]

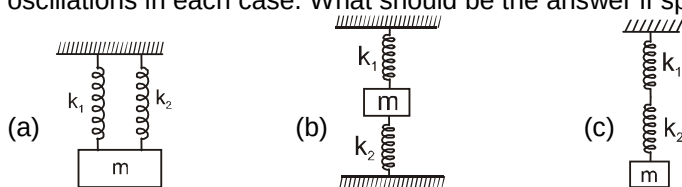
#### Section (C) : Spring mass system

- C-1.** A spring mass system has a time period of 2 second. What should be the spring constant of spring if the mass of the block is 10 grams ?
- C-2.** A body of mass 2 kg suspended through a vertical spring executes simple harmonic motion of period 4s. If the oscillations are stopped and the body hangs in equilibrium, find the potential energy stored in the spring.
- C-3.** A vertical spring-mass system with lower end of spring is fixed, made to undergo small oscillations. If the spring is stretched by 25cm, energy stored in the spring is 5J. Find the mass of the block if it makes 5 oscillations each second.
- C-4.** A spring mass system is shown in figure, spring is initially unstretched. A man starts pulling the block with constant force  $F$ . Find  
(a) The amplitude and the time period of motion of the block  
(b) The K.E. of the block at mean position  
(c) The energy stored in the spring when the block passes through the mean position

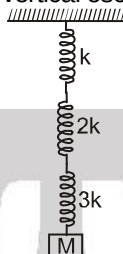




- C-5.** Three spring mass systems are shown in figure. Assuming gravity free space, find the time period of oscillations in each case. What should be the answer if space is not gravity free ?



- C-6.** Spring mass system is shown in figure. Find the elastic potential energy stored in each spring when block is at its mean position. Also find the time period of vertical oscillations. The system is in vertical plane.



### Section (D) : Simple Pendulum

- D-1.** Find the length of seconds pendulum at a place where  $g = \pi^2 \text{ m/s}^2$ .
- D-2.** Instantaneous angle (in radian) between string of a simple pendulum and vertical is given by  $\theta = \frac{\pi}{180} \sin 2\pi t$ . Find the length of the pendulum if  $g = \pi^2 \text{ m/s}^2$
- D-3.** A pendulum clock is accurate at a place where  $g = 9.8 \text{ m/s}^2$ . Find the value of  $g$  at another place where clock becomes slow by 24 seconds in a day (24 hrs).
- D-4.** A pendulum is suspended in a lift and its period of oscillation is  $T_0$  when the lift is stationary.
- What will be the period  $T$  of oscillation of pendulum, if the lift begins to accelerate downwards with an acceleration equal to  $\frac{3g}{4}$  ?
  - What must be the acceleration of the lift for the period of oscillation of the pendulum to be  $\frac{T_0}{2}$  ?

### Section (E) : Compound Pendulum & Torsional Pendulum

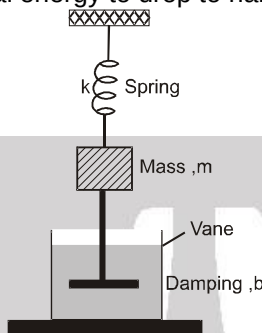
- E-1.** Compound pendulums are made of :
- A rod of length  $\ell$  suspended through a point located at distance  $\ell/4$  from centre of rod.
  - A ring of mass  $m$  and radius  $r$  suspended through a point on its periphery.
  - A uniform square plate of edge  $a$  suspended through a corner.
  - A uniform disc of mass  $m$  and radius  $r$  suspended through a point  $r/2$  away from the centre.
- Find the time period in each case.
- E-2.** Two compound pendulums are made of :
- A disc of radius  $r$  and
  - A uniform rod of length  $L$ . Find the minimum possible time period and distance between centre and point of suspension in each case.

### Section (F) : Superposition of SHM

- F-1.** A particle is subjected to two SHM's simultaneously  $X_1 = a_1 \sin \omega t$  and  $X_2 = a_2 \sin(\omega t + \phi)$  Where  $a_1 = 3.0 \text{ cm}$ ,  $a_2 = 4.0 \text{ cm}$ . Find resultant amplitude if the phase difference  $\phi$  has values (a)  $0^\circ$  (b)  $60^\circ$  (c)  $90^\circ$ .
- F-2.** A particle is subjected to three SHM's in same direction simultaneously each having equal amplitude  $a$  and equal time period. The phase of the second motion is  $30^\circ$  ahead of the first and the phase of the third motion is  $30^\circ$  ahead of the second. Find the amplitude of the resultant motion.
- F-3.** A particle simultaneously participates in two mutually perpendicular oscillations  $x = \sin \pi t$  &  $y = 2 \cos 2 \pi t$ . Write the equation of trajectory of the particle.

**Section (G) : For JEE-Main**

- G-1.** In forced oscillation of a particle, the amplitude is maximum for a frequency  $\omega_1$  of the force, while the energy is maximum for a frequency  $\omega_2$  of the force. What is the relation between  $\omega_1$  and  $\omega_2$  ?
- G-2.** For the damped oscillator shown in Figure, the mass of the block is 200 g,  $k = 80 \text{ Nm}^{-1}$  and the damping constant  $b$  is  $40 \text{ gs}^{-1}$ . Calculate
- The period of oscillation,
  - Time taken for its amplitude of vibrations to drop to half of its initial value
  - The time for the mechanical energy to drop to half initial value.

**PART - II : ONLY ONE OPTION CORRECT TYPE****Section (A) : Equation of SHM**

- A-1.** According to a scientists, he applied a force  $F = -cx^{1/3}$  on a particle and the particle is performing SHM. No other force acted on the particle. He refuses to tell whether  $c$  is a constant or not. Assume that he had worked only with positive  $x$  then :
- as  $x$  increases  $c$  also increases
  - as  $x$  increases  $c$  decreases
  - as  $x$  increases  $c$  remains constant
  - the motion cannot be SHM
- A-2.** A particle performing SHM takes time equal to  $T$  (time period of SHM) in consecutive appearances at a particular point. This point is :
- An extreme position
  - The mean position
  - Between positive extreme and mean position
  - Between negative extreme and mean position
- A-3.** A particle executing linear SHM. Its time period is equal to the smallest time interval in which particle acquires a particular velocity  $\vec{v}$ , the magnitude of  $\vec{v}$  may be :
- Zero
  - $V_{\max}$
  - $\frac{V_{\max}}{2}$
  - $\frac{V_{\max}}{\sqrt{2}}$
- A-4.** If  $\vec{F}$  is force vector,  $\vec{v}$  is velocity vector,  $\vec{a}$  vector is acceleration vector and  $\vec{r}$  vector is displacement vector with respect to mean position than which of the following quantities are always non-negative in a simple harmonic motion along a straight line?
- $\vec{F} \cdot \vec{a}$
  - $\vec{v} \cdot \vec{r}$
  - $\vec{a} \cdot \vec{r}$
  - $\vec{F} \cdot \vec{r}$
- A-5.** Two SHM's are represented by  $y = a \sin(\omega t - \phi)$  and  $y = b \cos(\omega t - \phi)$ . The phase difference between the two is :
- $\frac{\pi}{2}$
  - $\frac{\pi}{4}$
  - $\frac{\pi}{6}$
  - $\frac{3\pi}{4}$
- A-6.** How long after the beginning of motion is the displacement of a harmonically oscillating particle equal to one half its amplitude if the period is 24s and particle starts from rest.
- 12s
  - 2s
  - 4s
  - 6s
- A-7.** The magnitude of average acceleration in half time period from equilibrium position in a simple harmonic motion is
- $\frac{2A\omega^2}{\pi}$
  - $\frac{A\omega^2}{2\pi}$
  - $\frac{A\omega^2}{\sqrt{2}\pi}$
  - Zero





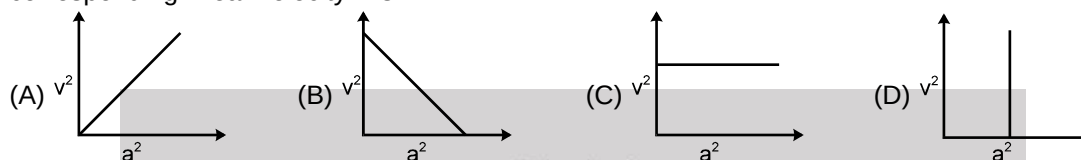
**A-8.** A particle performing SHM on the y axis according to equation  $y = A + B \sin \omega t$ . Its amplitude is :

- (A) A (B) B (C)  $A + B$  (D)  $\sqrt{A^2 + B^2}$

**A-9.** Two particles execute S.H.M. of same amplitude and frequency along the same straight line from same mean position. They cross one another without collision, when going in opposite directions, each time their displacement from mean position is half of their amplitudes. The phase-difference between them is

- (A)  $0^\circ$  (B)  $120^\circ$  (C)  $180^\circ$  (D)  $135^\circ$

**A-10.** A mass M is performing linear simple harmonic motion, then correct graph for acceleration a and corresponding linear velocity v is



### Section (B) : Energy

**B-1.** A body executing SHM passes through its equilibrium. At this instant, it has

- (A) maximum potential energy (B) maximum kinetic energy  
(C) minimum kinetic energy (D) maximum acceleration

**B-2.** The K.E. and P.E of a particle executing SHM with amplitude A will be equal when its displacement is

- (A)  $\sqrt{2}A$  (B)  $\frac{A}{2}$  (C)  $\frac{A}{\sqrt{2}}$  (D)  $\frac{\sqrt{2}}{3}A$

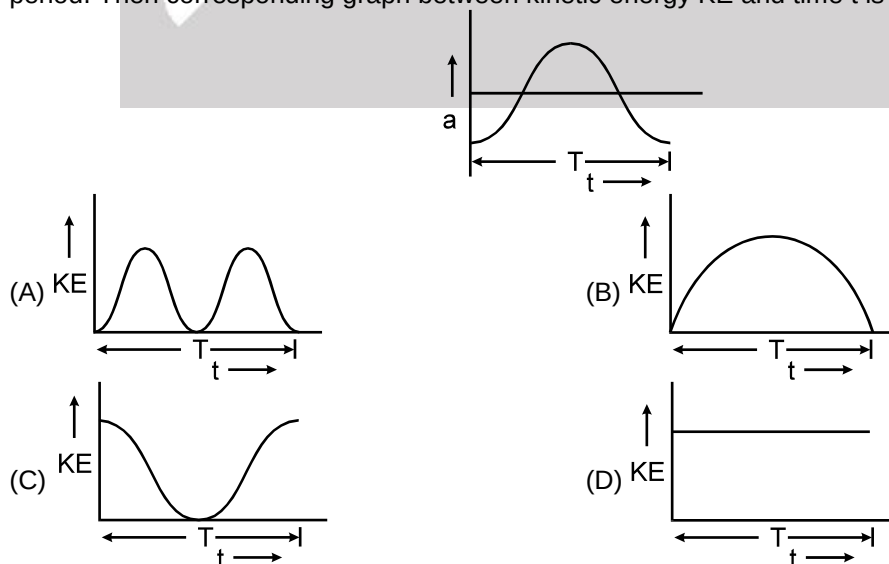
**B-3.** A point particle of mass 0.1 kg is executing S.H.M. of amplitude of 0.1 m. When the particle passes through the mean position, its kinetic energy is  $8 \times 10^{-3} \text{ J}$ . The equation of motion of this particle when the initial phase of oscillation is  $45^\circ$  can be given by [I.I.T. 1991]

- (A)  $0.1 \cos\left(4t + \frac{\pi}{4}\right)$  (B)  $0.1 \sin\left(4t + \frac{\pi}{4}\right)$  (C)  $0.4 \sin\left(t + \frac{\pi}{4}\right)$  (D)  $0.2 \sin\left(\frac{\pi}{2} + 2t\right)$

**B-4.** For a particle performing SHM :

- (A) The kinetic energy is never equal to the potential energy  
(B) the kinetic energy is always equal to the potential energy  
(C) The average kinetic energy in one time period is equal to the average potential in this period  
(D) The average kinetic energy in any time interval is equal to average potential energy in that interval

**B-5.** Acceleration a versus time t graph of a body in SHM is given by a curve shown below. T is the time period. Then corresponding graph between kinetic energy KE and time t is correctly represented by





- B-6.** A particle performs S.H.M. of amplitude  $A$  along a straight line. When it is at a distance  $\frac{\sqrt{3}}{2}A$  from mean position, its kinetic energy gets increased by an amount  $\frac{1}{2}m\omega^2 A^2$  due to an impulsive force.

Then its new amplitude becomes:

- (A)  $\frac{\sqrt{5}}{2}A$  (B)  $\frac{\sqrt{3}}{2}A$  (C)  $\sqrt{2}A$  (D)  $\sqrt{5}A$

### Section (C) : Spring mass system

- C-1.** Two spring mass systems have equal mass and spring constant  $k_1$  and  $k_2$ . If the maximum velocities in two systems are equal then ratio of amplitude of 1st to that of 2nd is :

- (A)  $\sqrt{k_1/k_2}$  (B)  $k_1/k_2$  (C)  $k_2/k_1$  (D)  $\sqrt{k_2/k_1}$

- C-2.** A toy car of mass  $m$  is having two similar rubber ribbons attached to it as shown in the figure. The force constant of each rubber ribbon is  $k$  and surface is frictionless. The car is displaced from mean position by  $x$  cm and released. At the mean position the ribbons are undeformed. Vibration period is



- (A)  $2\pi\sqrt{\frac{m(2k)}{k^2}}$  (B)  $\frac{1}{2\pi}\sqrt{\frac{m(2k)}{k^2}}$  (C)  $2\pi\sqrt{\frac{m}{k}}$  (D)  $2\pi\sqrt{\frac{m}{k+k}}$

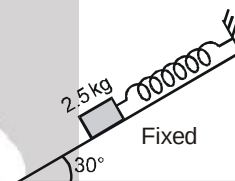
- C-3.** A mass of 1 kg attached to the bottom of a spring has a certain frequency of vibration. The following mass has to be added to it in order to reduce the frequency by half : [REE - 1988]

- (A) 1 kg (B) 2 kg (C) 3 kg (D) 4 kg

- C-4.** A ball of mass  $m$  kg hangs from a spring of spring constant  $k$ . The ball oscillates with a period of  $T$  seconds. If the ball is removed, the spring is shortened (with respect to length in mean position) by

- (A)  $\frac{gT^2}{(2\pi)^2}$  metre (B)  $\frac{3T^2g}{(2\pi)^2}$  metre (C)  $\frac{Tm}{k}$  metre (D)  $\frac{Tk}{m}$  metre

- C-5.** A smooth inclined plane having angle of inclination  $30^\circ$  with horizontal has a mass 2.5 kg held by a spring which is fixed at the upper end as shown in figure. If the mass is taken 2.5 cm up along the surface of the inclined plane, the tension in the spring reduces to zero. If the mass is then released, the angular frequency of oscillation in radian per second is

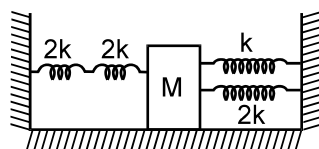


- (A) 0.707 (B) 7.07 (C) 1.414 (D) 14.14

- C-6.** A particle executes simple harmonic motion under the restoring force provided by a spring. The time period is  $T$ . If the spring is divided in two equal parts and one part is used to continue the simple harmonic motion, the time period will

- (A) remain  $T$  (B) become  $2T$  (C) become  $T/2$  (D) become  $T/\sqrt{2}$

- C-7.** Four massless springs whose force constants are  $2k$ ,  $2k$ ,  $k$  and  $2k$  respectively are attached to a mass  $M$  kept on a frictionless plane (as shown in figure). If the mass  $M$  is displaced in the horizontal direction, then the frequency of the system. [JEE 1990]



- (A)  $\frac{1}{2\pi}\sqrt{\frac{k}{4M}}$  (B)  $\frac{1}{2\pi}\sqrt{\frac{4k}{M}}$  (C)  $\frac{1}{2\pi}\sqrt{\frac{k}{7M}}$  (D)  $\frac{1}{2\pi}\sqrt{\frac{7k}{M}}$

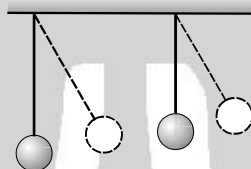




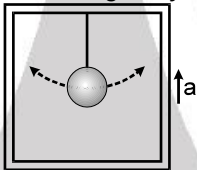
- C-8.** The total mechanical energy of a particle of mass  $m$  executing SHM with the help of a spring is  $E = (1/2)m\omega^2 A^2$ . If the particle is replaced by another particle of mass  $m/2$  while the amplitude  $A$  remains same. New mechanical energy will be :
- (A)  $\sqrt{2} E$  (B)  $2E$  (C)  $E/2$  (D)  $E$

### Section (D) : Simple Pendulum

- D-1.** Two pendulums begin to swing simultaneously. The first pendulum makes 9 full oscillations when the other makes 7. Find the ratio of length of the two pendulums.
- (A)  $\frac{49}{81}$  (B)  $\frac{7}{9}$  (C)  $\frac{50}{81}$  (D)  $\frac{1}{2}$
- D-2.** Two pendulums at rest start swinging together. Their lengths are respectively 1.44 m and 1 m. They will again start swinging in same phase together after (of longer pendulum) :



- (A) 1 vibration (B) 3 vibrations (C) 4 vibrations (D) 5 vibrations
- D-3.** A scientist measures the time period of a simple pendulum as  $T$  in a lift at rest. If the lift moves up with acceleration as one fourth of the acceleration of gravity, the new time period is



- (A)  $\frac{T}{4}$  (B)  $4T$  (C)  $\frac{2}{\sqrt{5}} T$  (D)  $\frac{\sqrt{5}}{2} T$
- D-4.** A simple pendulum has some time period  $T$ . What will be the percentage change in its time period if its amplitude is decreased by 5%?
- (A) 6 % (B) 3 % (C) 1.5 % (D) 0 %
- D-5.** A simple pendulum with length  $\ell$  and bob of mass  $m$  executes SHM of small amplitude  $A$ . The maximum tension in the string will be
- (A)  $mg(1 + A/\ell)$  (B)  $mg(1 + A/\ell)^2$  (C)  $mg[1 + (A/\ell)^2]$  (D)  $2 mg$

### Section (E) : Compound Pendulum & Torsional Pendulum

- E-1.** A 25 kg uniform solid sphere with a 20 cm radius is suspended by a vertical wire such that the point of suspension is vertically above the centre of the sphere. A torque of 0.10 N-m is required to rotate the sphere through an angle of 1.0 rad and then the orientation is maintained. If the sphere is then released, its time period of the oscillation will be :
- (A)  $\pi$  second (B)  $\sqrt{2} \pi$  second (C)  $2\pi$  second (D)  $4\pi$  second
- E-2.** A metre stick swinging about its one end oscillates with frequency  $f_0$ . If the bottom half of the stick was cut off, then its new oscillation frequency will be :
- (A)  $f_0$  (B)  $\sqrt{2} f_0$  (C)  $2f_0$  (D)  $2 f_0$

### Section (F) : Superposition of SHM

- F-1.** When two mutually perpendicular simple harmonic motions of same frequency, amplitude and phase are superimposed
- (A) the resulting motion is uniform circular motion.
- (B) the resulting motion is a linear simple harmonic motion along a straight line inclined equally to the straight lines of motion of component ones.
- (C) the resulting motion is an elliptical motion, symmetrical about the lines of motion of the components.
- (D) the two S.H.M. will cancel each other.







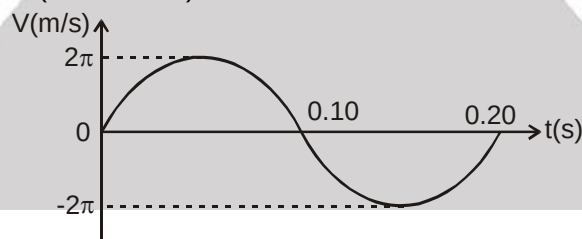
- F-2.** The position of a particle in motion is given by  $y = C\sin\omega t + D\cos\omega t$  w.r.t. origin. Then motion of the particle is:
- (A) SHM with amplitude  $C+D$  (B) SHM with amplitude  $\sqrt{C^2 + D^2}$   
 (C) SHM with amplitude  $\frac{(C+D)}{2}$  (D) not SHM
- F-3.** A simple harmonic motion is given by  $y = 5 (\sin 3\pi t + \sqrt{3} \cos 3\pi t)$ . What is the amplitude of motion if  $y$  is in m ?  
 (A) 100 cm (B) 5 m (C) 200 cm (D) 1000 cm
- F-4.** The position vector of a particle moving in x-y plane is given by  $\vec{r} = (A \sin\omega t) \hat{i} + (A \cos\omega t) \hat{j}$  then motion of the particle is :  
 (A) SHM (B) on a circle (C) on a straight line (D) with constant acceleration

### Section (G) : For JEE Main

- G-1.** When an oscillator completes 100 oscillations its amplitude reduced to  $\frac{1}{3}$  of initial value. What will be its amplitude, when it completes 200 oscillations : [AIPMT 2002]  
 (A)  $\frac{1}{8}$  (B)  $\frac{2}{3}$  (C)  $\frac{1}{6}$  (D)  $\frac{1}{9}$
- G-2.** The damping force on an oscillator is directly proportional to the velocity. The units of the constant of proportionality are :  
 (A)  $\text{kgms}^{-1}$  (B)  $\text{kgms}^{-2}$  (C)  $\text{kgs}^{-1}$  (D) kgs

## PART - III : MATCH THE COLUMN

- 1.** A simple harmonic oscillator consists of a block attached to a spring with  $k = 200 \text{ N/m}$ . The block slides on a frictionless horizontal surface, with equilibrium point  $x = 0$ . A graph of the block's velocity  $v$  as a function of time  $t$  is shown. Correctly match the required information in the left column with the values given in the right column. (Use  $\pi^2 = 10$ )



### Left Column

- (A) The block's mass in kg  
 (B) The block's displacement at  $t = 0$  in metres  
 (C) The block's acceleration at  $t = 0.10 \text{ s}$  in  $\text{m/s}^2$   
 (D) The block's maximum kinetic energy in Joule

### Right Column

- (p)  $-0.20$   
 (q)  $-200$   
 (r)  $0.20$   
 (s)  $4.0$





2. In the column-I, a system is described in each option and corresponding time period is given in the column-II. Suitably match them.

## Column-I

- (A) A simple pendulum of length ' $\ell$ ' oscillating

with small amplitude in a lift moving down with retardation  $g/2$ .

- (B) A block attached to an end of a vertical

spring, whose other end is fixed to the ceiling of a stationary lift, stretches the spring by length ' $\ell$ ' in equilibrium. It's time period when lift moves up with an acceleration  $g/2$  is

- (C) The time period of small oscillation of a

uniform rod of length ' $\ell$ ' smoothly hinged at one end. The rod oscillates in vertical plane.

- (D) A cubical block of edge ' $\ell$ ' and specific

gravity  $1/2$  is in equilibrium with some volume inside water filled in a large fixed container. Neglect viscous forces and surface tension. The time period of small oscillations of the block in vertical direction is

## Column-II

(p)  $T = 2\pi \sqrt{\frac{2\ell}{3g}}$

(q)  $T = 2\pi \sqrt{\frac{\ell}{g}}$

(r)  $T = 2\pi \sqrt{\frac{2\ell}{g}}$

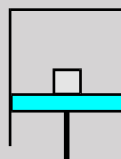
(s)  $T = 2\pi \sqrt{\frac{\ell}{2g}}$

## Exercise-2

Marked Questions can be used as Revision Questions.

### PART - I : ONLY ONE OPTION CORRECT TYPE

1. A block of mass  $m$  is resting on a piston as shown in figure which is moving vertically with a SHM of period 1 s. The minimum amplitude of motion at which the block and piston separate is :



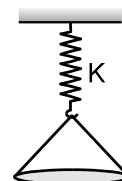
- (A) 0.25 m (B) 0.52 m (C) 2.5 m (D) 0.15 m

2. The potential energy of a particle of mass ' $m$ ' situated in a unidimensional potential field varies as  $U(x) = U_0 [1 - \cos ax]$ , where  $U_0$  and  $a$  are constants. The time period of small oscillations of the particle about the mean position :

- (A)  $2\pi \sqrt{\frac{m}{aU_0}}$  (B)  $2\pi \sqrt{\frac{am}{U_0}}$  (C)  $2\pi \sqrt{\frac{m}{a^2U_0}}$  (D)  $2\pi \sqrt{\frac{a^2m}{U_0}}$

3. A solid ball of mass  $m$  is made to fall from a height  $H$  on a pan suspended through a spring of spring constant  $K$  as shown in figure. If the ball does not rebound and the pan is massless, then amplitude of oscillation is

- (A)  $\frac{mg}{K}$  (B)  $\frac{mg}{K} \left( 1 + \frac{2HK}{mg} \right)^{1/2}$   
 (C)  $\frac{mg}{K} + \left( \frac{2HK}{mg} \right)^{1/2}$  (D)  $\frac{mg}{K} \left[ 1 + \left( 1 + \frac{2HK}{mg} \right)^{1/2} \right]$

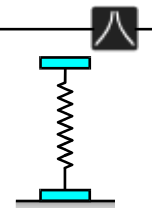




## Simple Harmonic Motion

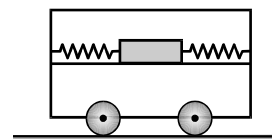
4. Two plates of same mass are attached rigidly to the two ends of a spring as shown in figure. One of the plates rests on a horizontal surface and the other results a compression  $y$  of the spring when it is in equilibrium state. The further minimum compression required, so that after the force causing compression is removed the lower plate is lifted off the surface, will be :

(A)  $0.5y$  (B)  $3y$  (C)  $2y$  (D)  $y$

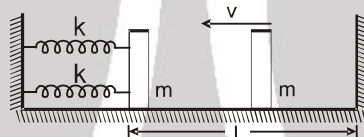


5. Two springs, each of spring constant  $k$ , are attached to a block of mass  $m$  as shown in the figure. The block can slide smoothly along a horizontal platform clamped to the opposite walls of the trolley of mass  $M$ . If the block is displaced by  $x$  cm and released, the period of oscillation is :

(A)  $T = 2\pi \sqrt{\frac{Mm}{2k}}$  (B)  $T = 2\pi \sqrt{\frac{(M+m)}{kmM}}$  (C)  $T = 2\pi \sqrt{\frac{mM}{2k(M+m)}}$  (D)  $T = 2\pi \frac{(M+m)^2}{k}$



6. The right block in figure moves at a speed  $V$  towards the left block placed in equilibrium. All the surfaces are smooth and all the collisions are elastic. Find the time period of periodic motion. Neglect the width of the blocks.



(A)  $\pi \sqrt{\frac{m}{2k}} + \frac{2L}{v}$  (B)  $\pi \sqrt{\frac{m}{2k}} + \frac{L}{v}$  (C)  $\pi \sqrt{\frac{m}{2k}} - \frac{L}{v}$  (D)  $\pi \sqrt{\frac{m}{k}} + \frac{L}{v}$

7. The bob in a simple pendulum of length  $\ell$  is released at  $t = 0$  from the position of small angular displacement  $\theta_0$ . Linear displacement of the bob at any time  $t$  from the mean position is given by

(A)  $\ell \theta_0 \cos \sqrt{\frac{g}{\ell}} t$  (B)  $\ell \sqrt{\frac{g}{\ell}} t \cos \theta_0$  (C)  $\ell g \sin \theta_0$  (D)  $\ell \theta_0 \sin \sqrt{\frac{g}{\ell}} t$

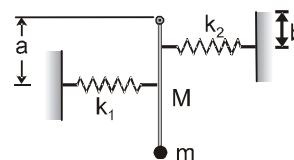
8. The period of small oscillations of a simple pendulum of length  $\ell$  if its point of suspension  $O$  moves with a constant acceleration  $\alpha = \alpha_1 \hat{i} - \alpha_2 \hat{j}$  with respect to earth is ( $\hat{i}$  and  $\hat{j}$  are unit vectors in horizontal and vertically upward directions respectively)

(A)  $T = 2\pi \sqrt{\frac{\ell}{\{(g - \alpha_2)^2 + \alpha_1^2\}^{1/2}}}$  (B)  $T = 2\pi \sqrt{\frac{\ell}{\{(g - \alpha_1)^2 + \alpha_2^2\}^{1/2}}}$   
(C)  $T = 2\pi \sqrt{\frac{\ell}{g}}$  (D)  $T = 2\pi \sqrt{\frac{\ell}{(g^2 + \alpha_1^2)^{1/2}}}$

9. A simple pendulum ; a physical pendulum; a torsional pendulum and a spring-mass system, each of same frequency are taken to the Moon. If frequencies are measured on the moon, which system or systems will have it unchanged ?

(A) spring-mass system and torsional pendulum. (B) only spring-mass system.  
(C) spring-mass system and physical pendulum. (D) None of these

10. A rod of mass  $M$  and length  $L$  is hinged at its one end and carries a particle of mass  $m$  at its lower end. A spring of force constant  $k_1$  is installed at distance  $a$  from the hinge and another of force constant  $k_2$  at a distance  $b$  as shown in the figure. If the whole arrangement rests on a smooth horizontal table top, the frequency of vibration is



(A)  $\frac{1}{2\pi} \sqrt{\frac{k_1 a^2 + k_2 b^2}{L^2(m + \frac{M}{3})}}$  (B)  $\frac{1}{2\pi} \sqrt{\frac{k_2 + k_1}{M + m}}$  (C)  $\frac{1}{2\pi} \sqrt{\frac{k_2 + k_1 \frac{a^2}{b^2}}{\frac{4M}{3} + m}}$  (D)  $\frac{1}{2\pi} \sqrt{\frac{k_1 + \frac{k_2 b^2}{a^2}}{\frac{4}{3}m + M}}$



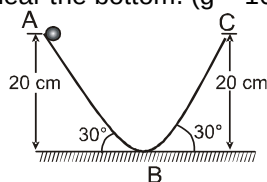
11. A particle moves along the X-axis according to the equation  $x = 10 \sin^3(\pi t)$ . The amplitudes and frequencies of component SHMs are  
 (A) amplitude 30/4, 10/4 ; frequencies 3/2, 1/2 (B) amplitude 30/4, 10/4 ; frequencies 1/2, 3/2  
 (C) amplitude 10, 10 ; frequencies 1/2, 1/2 (D) amplitude 30/4, 10 ; frequencies 3/2, 2
12. The amplitude of a particle due to superposition of following S.H.Ms. Along the same line is  
 $X_1 = 2 \sin 50\pi t$  ;  $X_2 = 10 \sin(50\pi t + 37^\circ)$   
 $X_3 = -4 \sin 50\pi t$  ;  $X_4 = -12 \cos 50\pi t$   
 (A)  $4\sqrt{2}$  (B) 4 (C)  $6\sqrt{2}$  (D) none of these
13. When a body is suspended from a fixed point by a spring, the angular frequency of its vertical oscillations is  $\omega_1$ . When a different spring is used, the angular frequency is  $\omega_2$ . The angular frequency of vertical oscillations when both the springs are used together in series is given by [Olympiad (Stage-1) 2017]  
 (A)  $\omega = \left[ \omega_1^2 + \omega_2^2 \right]^{\frac{1}{2}}$  (B)  $\omega = \left[ \frac{\omega_1^2 + \omega_2^2}{2} \right]^{\frac{1}{2}}$  (C)  $\omega = \left[ \frac{\omega_1^2 \omega_2^2}{(\omega_1^2 + \omega_2^2)} \right]^{\frac{1}{2}}$  (D)  $\omega = \left[ \frac{\omega_1^2 \omega_2^2}{2(\omega_1^2 + \omega_2^2)} \right]^{\frac{1}{2}}$
14. A particle performs simple harmonic motion at a frequency  $f$ . The frequency at which its kinetic energy varies is : [Olympiad (Stage-1) 2017]  
 (A)  $f$  (B)  $2f$  (C)  $4f$  (D)  $f/2$
15. A particle rests in equilibrium under two forces of repulsion whose centres are at distance of  $a$  and  $b$  from the particle. The forces vary as the cube of the distance. The forces per unit mass are  $k$  and  $k'$  respectively. If the particle be slightly displaced towards one of them the motion is simple harmonic with the time period equal to [Olympiad (Stage-1) 2017]  
 (A)  $\frac{2\pi}{\sqrt{3\left(\frac{k}{a^3} + \frac{k'}{b^3}\right)}}$  (B)  $\frac{2\pi}{\sqrt{\left(\frac{k}{a^3} + \frac{k'}{b^3}\right)}}$  (C)  $\frac{2\pi}{\sqrt{\left(\frac{k}{a^4} + \frac{k'}{b^4}\right)}}$  (D)  $\frac{2\pi}{\sqrt{3\left(\frac{k}{a} + \frac{k'}{b}\right)}}$

## PART - II : SINGLE AND DOUBLE VALUE INTEGER TYPE

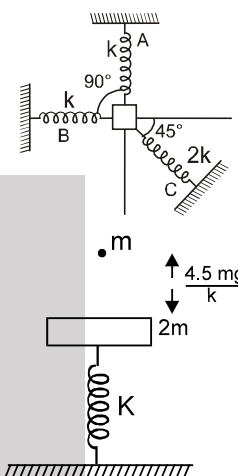
1. Two particles A and B are performing SHM along x and y-axis respectively with equal amplitude and frequency of 2 cm and 1 Hz respectively. Equilibrium positions of the particles A and B are at the co-ordinates (3, 0) and (0, 4) respectively. At  $t = 0$ , B is at its equilibrium position and moving towards the origin, while A is nearest to the origin and moving away from the origin. If the maximum and minimum distances between A and B is  $s_1$  and  $s_2$  then find  $s_1 + s_2$  (in cm).
2. Two particles P and Q describe S.H.M. of same amplitude  $a$ , same frequency  $f$  along the same straight line from the same mean position. The maximum distance between the two particles is  $a\sqrt{2}$ . If the initial phase difference between the particles is  $\pi/N$  then find  $N$ .
3. A street car moves rectilinearly from station A (here car stops) to the next station B (here also car stops) with an acceleration varying according to the law  $f = a - bx$ , where  $a$  and  $b$  are positive constants and  $x$  is the distance from station A. If the maximum distance between the two stations is  $x = \frac{Na}{b}$  then find  $N$ .
4. A particle is oscillating in a straight line about a centre O, with a force directed towards O. When at a distance ' $x$ ' from O, the force is  $mn^2x$  where ' $m$ ' is the mass and ' $n$ ' is a constant. The amplitude is  $a = 15$  cm. When at a distance  $\sqrt{3}\frac{a}{2}$  from O the particle receives a blow in the direction of motion which generates an extra velocity  $na$ . If the velocity is away from O at the time of blow and the new amplitude becomes  $k\sqrt{3}$  cm, then find  $k$ .
5. Two particles  $P_1$  and  $P_2$  are performing SHM along the same line about the same mean position. Initially they are at their positive extreme positions. If the time period of each particle is 12 sec and the difference of their amplitudes is 12 cm then find the minimum time after which the separation between the particles become 6 cm.



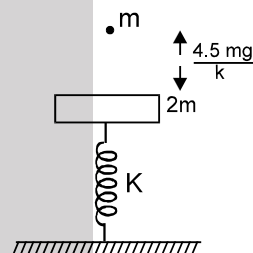
6. Assuming all the surfaces to be smooth, if the time period of motion of the ball is  $N \times 10^{-1}$  sec then find N. Neglect the small effect of bend near the bottom. ( $g = 10 \text{ m/s}^2$ )



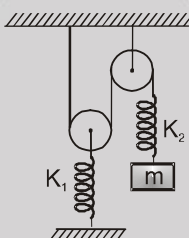
7. A block of mass  $m$  is attached to three springs A, B and C having force constants  $k$ ,  $k$  and  $2k$  respectively as shown in figure. If the block is slightly pushed against spring C. If the angular frequency of oscillations is  $\sqrt{\frac{Nk}{m}}$ , then find N. The system is placed on horizontal smooth surface.



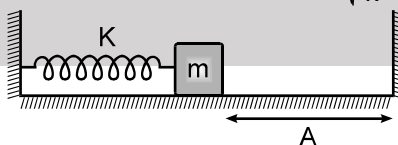
8. In the figure shown mass  $2m$  is at rest and in equilibrium. A particle of mass  $m$  is released from height  $\frac{4.5mg}{k}$  from plate. The particle sticks to the plate. Neglecting the duration of collision. Starting from the time when the particle sticks to plate to the time when the spring is in maximum compression for the first time is  $2\pi\sqrt{\frac{m}{ak}}$  then find  $a$ .



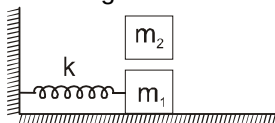
9. For given spring mass system, If the time period of small oscillations of block about its mean position is  $\pi\sqrt{\frac{nm}{K}}$ , then find  $n$ . Assume ideal conditions. The system is in vertical plane and take  $K_1 = 2K$ ,  $K_2 = K$ .



10. In the figure shown the spring is relaxed and mass  $m$  is attached to the spring. The spring is compressed by  $2A$  and released at  $t = 0$ . Mass  $m$  collides with the wall and loses two third of its kinetic energy and returns. Starting from  $t = 0$ , find the time taken by it to come back to rest again (instant at which spring is again under maximum compression). Take  $\sqrt{\frac{m}{k}} = \frac{12}{\pi}$



11. A block of mass  $4 \text{ kg}$  attached with spring of spring constant  $100 \text{ N/m}$  is executing SHM of amplitude  $0.1 \text{ m}$  on smooth horizontal surface as shown in figure. If another block of mass  $5 \text{ kg}$  is gently placed on it, at the instant it passes through the mean position and new amplitude of motion is  $n^{-1}$  meter then find  $n$ . Assuming that two blocks always move together.

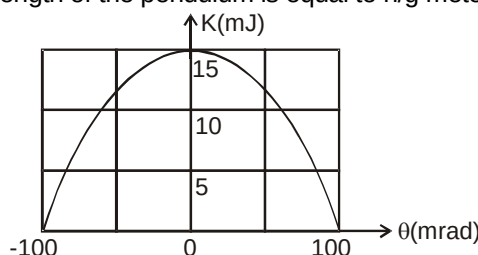


12. The period of oscillation of a simple pendulum of length  $L$  suspended from the roof of a vehicle which moves without friction down on inclined plane of inclination  $\alpha = 60^\circ$  is given by  $\pi\sqrt{\frac{XL}{g}}$  then find  $X$ .

[I.I.T. (Scr.) 2000, 1/35]



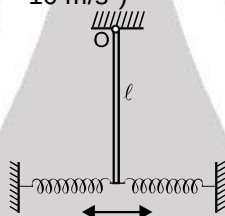
13. Figure shows the kinetic energy  $K$  of a simple pendulum versus its angle  $\theta$  from the vertical. The pendulum bob has mass  $0.2 \text{ kg}$ . If the length of the pendulum is equal to  $n/g$  meter, then find  $n$  ( $g = 10 \text{ m/s}^2$ ).



14. The bob of a simple pendulum executes simple harmonic motion in water with a period ' $t$ ', while the period of oscillation of the bob is  $t_0$  in air. Neglecting frictional force of water and given that the density of the bob is  $(4/3) \times 1000 \text{ kg/m}^3$ . Find  $\frac{t}{t_0}$ . [AIEEE 2004]

15. A solid sphere of radius  $R$  is half submerged in a liquid of density  $\rho$ . The sphere is slightly pushed down and released, If the frequency of small oscillations is  $\sqrt{\frac{3}{nR}}$ , then find  $n$ . Take  $\pi = \sqrt{g}$ . [JEE (Mains) 2004, 2/60]

16. If the angular frequency of small oscillations of a thin uniform vertical rod of mass  $m$  and length  $\ell$  hinged at the point  $O$  (Fig.) is  $\sqrt{\frac{n}{\ell}}$ , then find  $n$ . The force constant for each spring is  $K/2$  and take  $K = \frac{2mg}{\ell}$ . The springs are of negligible mass. ( $g = 10 \text{ m/s}^2$ )



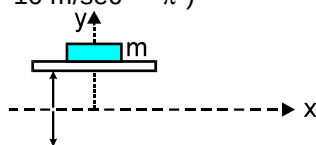
### PART - III : ONE OR MORE THAN ONE OPTIONS CORRECT TYPE

1. A particle moves on the X-axis according to the equation  $x = x_0 \sin^2 \omega t$ . The motion is simple harmonic  
(A) with amplitude  $x_0/2$  (B) with amplitude  $2x_0$  (C) with time period  $\frac{2\pi}{\omega}$  (D) with time period  $\frac{\pi}{\omega}$
2. Which of the following functions represent SHM?  
(A)  $\sin 2\omega t$  (B)  $\sin^2 \omega t$  (C)  $\sin \omega t + 2 \cos \omega t$  (D)  $\sin \omega t + \cos 2\omega t$
3. The speed  $v$  of a particle moving along a straight line, when it is at a distance ( $x$ ) from a fixed point of the line is given by  $v^2 = 108 - 9x^2$  (assuming mean position to have zero phase constant) (all quantities are in cgs units) :  
(A) the motion is uniformly accelerated along the straight line  
(B) the magnitude of the acceleration at a distance  $3 \text{ cm}$  from the fixed point is  $27 \text{ cm/s}^2$   
(C) the motion is simple harmonic about the given fixed point.  
(D) the maximum displacement from the fixed point is  $4 \text{ cm}$ .
4. A horizontal plank has a rectangular block placed on it. The plank starts oscillating vertically and simple harmonically with an amplitude of  $40 \text{ cm}$ . The block just loses contact with the plank when the latter is at momentary rest. Then :  
(A) the period of oscillation is  $\left(\frac{2\pi}{5}\right)$  seconds  
(B) the block weighs double its weight, when the plank is at one of the positions of momentary rest.  
(C) the block weighs  $0.5$  times its weight on the plank halfway up  
(D) the block weighs  $1.5$  times its weight on the plank halfway down  
(E) the block weighs its true weight on the plank when the latter moves fastest

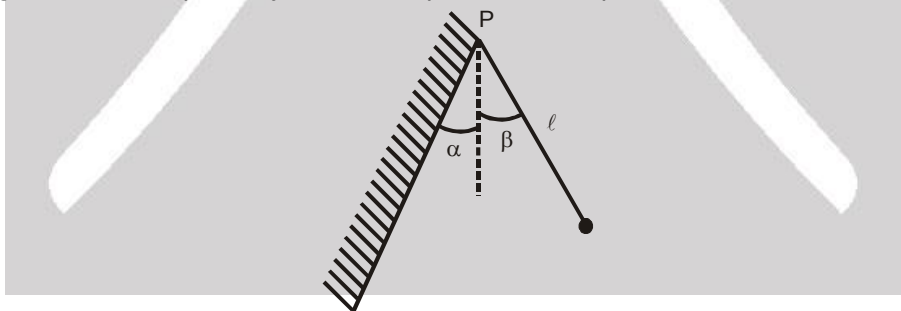




5. As shown in figure a horizontal platform with a mass  $m$  placed on it is executing SHM along  $y$ -axis. If the amplitude of oscillation is 2.5 cm, the minimum period of the motion for the mass not to be detached from the platform is : ( $g = 10 \text{ m/sec}^2 = \pi^2$ )



- (A)  $\frac{10}{\pi} \text{ s}$  (B)  $\frac{\pi}{10} \text{ s}$  (C)  $\frac{\pi}{\sqrt{10}} \text{ s}$  (D)  $\frac{1}{\sqrt{10}} \text{ s}$ .
6. For a body executing SHM with amplitude  $A$ , time period  $T$ , maximum velocity  $v_{\max}$  and phase constant zero, which of the following statements are correct for  $0 \leq t \leq \frac{T}{4}$  ( $y$  is displacement from mean position) ?
- (A) At  $y = (A/2)$ ,  $v > (v_{\max}/2)$  (B) for  $v = (v_{\max}/2)$ ,  $y > (A/2)$   
 (C) For  $t = (T/8)$ ,  $y > (A/2)$  (D) For  $y = (A/2)$ ,  $t < (T/8)$
7. The potential energy of a particle of mass 0.1 kg, moving along the  $x$ -axis, is given by  $U = 5x(x - 4) \text{ J}$ , where  $x$  is in meters. It can be concluded that
- (A) the particle is acted upon by a constant force (B) the speed of the particle is maximum at  $x = 2 \text{ m}$   
 (C) the particle executes SHM (D) the period of oscillation of the particle is  $(\pi/5) \text{ sec}$
8. A particle free to move along the  $x$ -axis has potential energy given by  $U(x) = k[1 - e^{-x^2}]$  for  $-\infty \leq x \leq +\infty$ , where  $k$  is a positive constant of appropriate dimensions. Then select the incorrect options : [JEE - 1999, 2/200]
- (A) at points away from the origin, the particle is in unstable equilibrium.  
 (B) for any finite non-zero value of  $x$ , there is a force directed away from the origin.  
 (C) if its total mechanical energy is  $k/2$ , it has its minimum kinetic energy at the origin.  
 (D) for small displacements from  $x = 0$ , the motion is simple harmonic.
9. A ball is hung vertically by a thread of length  $\ell$  from a point 'P' of an inclined wall that makes an angle ' $\alpha$ ' with the vertical. The thread with the ball is then deviated through a small angle ' $\beta$ ' ( $\beta > \alpha$ ) and set free. Assuming the wall to be perfectly elastic, the period of such pendulum is/are



- (A)  $2\sqrt{\frac{\ell}{g}} \left[ \sin^{-1} \left( \frac{\alpha}{\beta} \right) \right]$  (B)  $2\sqrt{\frac{\ell}{g}} \left[ \frac{\pi}{2} + \sin^{-1} \left( \frac{\alpha}{\beta} \right) \right]$   
 (C)  $2\sqrt{\frac{\ell}{g}} \left[ \cos^{-1} \left( \frac{\alpha}{\beta} \right) \right]$  (D)  $2\sqrt{\frac{\ell}{g}} \left[ \cos^{-1} \left( -\frac{\alpha}{\beta} \right) \right]$
10. If a SHM is given by  $y = (\sin \omega t + \cos \omega t) \text{ m}$ , which of the following statements are true?
- (A) The amplitude is 1m (B) The amplitude is  $\sqrt{2} \text{ m}$   
 (C) Time is considered from  $y = 1 \text{ m}$  (D) Time is considered from  $y = 0 \text{ m}$



11. The position of a particle at time  $t$  moving in  $x$ - $y$  plane is given by  $\vec{r} = (\hat{i} + 2\hat{j}) A \cos \omega t$ . Then, the motion of the particle is :  
 (A) on a straight line (B) on an ellipse (C) periodic (D) SHM
12. Three simple harmonic motions in the same direction having the same amplitude  $a$  and same period are superposed. If each differs in phase from the next by  $45^\circ$ , then, [IIT- 1999, 3/100]  
 (A) the resultant amplitude is  $(1+\sqrt{2})a$   
 (B) the phase of the resultant motion relative to the first is  $90^\circ$ .  
 (C) the energy associated with the resulting motion is  $(3+2\sqrt{2})$  times the energy associated with any single motion.  
 (D) the resulting motion is not simple harmonic.

## PART - IV : COMPREHENSION

### Comprehension-1

A 2kg block hangs without vibrating at the bottom end of a spring with a force constant of 400 N/m. The top end of the spring is attached to the ceiling of an elevator car. The car is rising with an upward acceleration of  $5 \text{ m/s}^2$  when the acceleration suddenly ceases at time  $t = 0$  and the car moves upward with constant speed. ( $g = 10 \text{ m/s}^2$ )

1. What is the angular frequency of oscillation of the block after the acceleration ceases?  
 (A)  $10\sqrt{2} \text{ rad/s}$  (B)  $20 \text{ rad/s}$  (C)  $20\sqrt{2} \text{ rad/s}$  (D)  $32 \text{ rad/s}$
2. The amplitude of the oscillations is  
 (A) 7.5 cm (B) 5 cm (C) 2.5 cm (D) 1 cm
3. The initial phase angle observed by an observer in the elevator, taking upward direction to be positive and positive extreme position to have  $\pi/2$  phase, is equal to  
 (A)  $-\pi/4 \text{ rad}$  (B)  $\pi/2 \text{ rad}$  (C)  $\pi \text{ rad}$  (D)  $3\pi/2 \text{ rad}$

### Comprehension-2

A particle of mass ' $m$ ' moves on a horizontal smooth line AB of length ' $a$ ' such that when particle is at any general point P on the line two forces act on it. A force  $\frac{mg(AP)}{a}$  towards A and another force  $\frac{2mg(BP)}{a}$  towards B.

4. Find its time period when released from rest from mid-point of line AB.  
 (A)  $T = 2\pi\sqrt{\frac{3a}{g}}$  (B)  $T = 2\pi\sqrt{\frac{a}{2g}}$  (C)  $T = 2\pi\sqrt{\frac{a}{g}}$  (D)  $T = 2\pi\sqrt{\frac{a}{3g}}$
5. Find the minimum distance of the particle from B during the motion.  
 (A)  $\frac{a}{6}$  (B)  $\frac{a}{4}$  (C)  $\frac{a}{3}$  (D)  $\frac{a}{8}$
6. If the force acting towards A stops acting when the particle is nearest to B then find the velocity with which it crosses point B.  
 (A)  $\frac{\sqrt{2ga}}{3}$  (B)  $\frac{\sqrt{2ga}}{6}$  (C)  $\frac{\sqrt{2ga}}{5}$  (D)  $\frac{\sqrt{ga}}{3}$

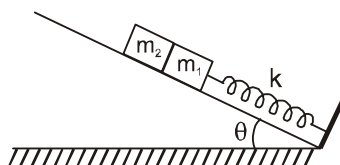






## Comprehension-3

Spring of spring constant  $k$  is attached with a block of mass  $m_1$  as shown in figure. Another block of mass  $m_2$  is placed against  $m_1$  and both masses lie on smooth incline plane.



7. Find the compression in the spring when the system is in equilibrium.
- (A)  $\frac{(m_1 + m_2)g \sin \theta}{2k}$  (B)  $\frac{(m_1 + m_2)g \sin \theta}{k}$
- (C)  $\frac{(m_1 + m_2)g}{k}$  (D)  $\frac{2(m_1 + m_2)g \sin \theta}{k}$
8. From the equilibrium position the blocks are pushed a further distance  $\frac{2}{k}(m_1 + m_2)g \sin \theta$  against the spring and released. Find the common speed of blocks when they separate.
- (A)  $\left( \sqrt{\frac{1}{3k}}(m_1 + m_2) \right) g \sin \theta$  (B)  $\left( \sqrt{\frac{2}{k}}(m_1 + m_2) \right) g \sin \theta$
- (C)  $\left( \sqrt{\frac{3}{k}}(m_1 + m_2) \right) g \sin \theta$  (D)  $\left( \sqrt{\frac{1}{k}}(m_1 + m_2) \right) g \sin \theta$

## Exercise-3

Marked Questions can be used as Revision Questions.

\* Marked Questions may have more than one correct option.

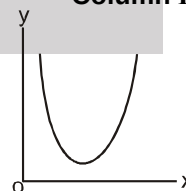
### PART - I : JEE (ADVANCED) / IIT-JEE PROBLEMS (PREVIOUS YEARS)

- 1\*. Function  $x = A \sin^2 \omega t + B \cos^2 \omega t + C \sin \omega t \cos \omega t$  represents SHM [JEE 2006, 5/184, -1]
- (A) for any value of  $A$ ,  $B$  and  $C$  (except  $C = 0$ ) (B) If  $A = -B$ ,  $C = 2B$ , amplitude =  $|B\sqrt{2}|$
- (C) If  $A = B$ ;  $C = 0$  (D) If  $A = B$ ;  $C = 2B$ , amplitude =  $|B|$
2. Column I gives a list of possible set of parameters measured in some experiments. The variations of the parameters in the form of graphs are shown in Column II. Match the set of parameters given in Column I with the graphs given in Column II. [IIT-JEE 2008, 6/163]

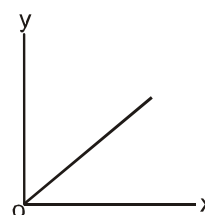
Column I

Column II

- (A) Potential energy of a simple pendulum (y-axis) as a function of displacement (x-axis)

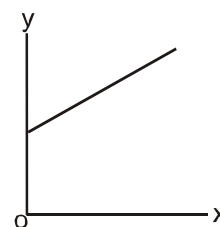


- (B) Displacement (y-axis) as a function of time (x-axis) for a one dimensional motion at zero or constant acceleration when the body is moving along the positive x-direction.

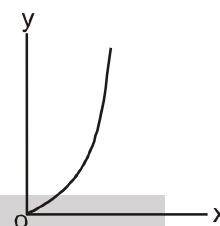




- (C) Range of a projectile (y-axis) as a function of its velocity (x-axis) when projected at a fixed angle.

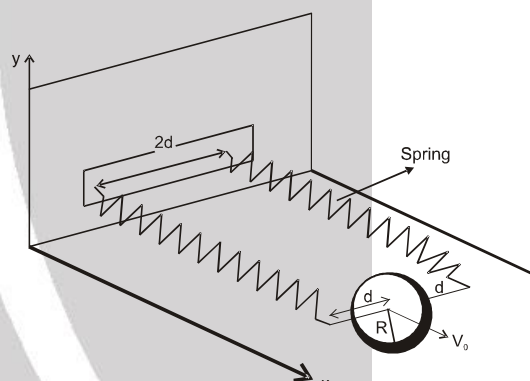


- (D) The square of the time period (y-axis) of a simple pendulum as a function of its length (x-axis).



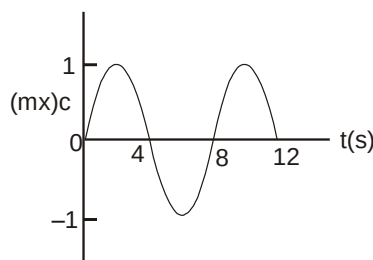
### Comprehension-1

A uniform thin cylindrical disk of mass  $M$  and radius  $R$  is attached to two identical massless springs of spring constant  $k$  which are fixed to the wall as shown in the figure. The springs are attached to the axle of the disk symmetrically on either side at a distance  $d$  from its centre. The axle is massless and both the springs and the axle are in a horizontal plane. The unstretched length of each spring is  $L$ . The disk is initially at its equilibrium position with its centre of mass (CM) at a distance  $L$  from the wall. The disk rolls without slipping with velocity  $\vec{V}_0 = V_0 \hat{i}$ . The coefficient of friction is  $\mu$ . Figure :



[JEE-2008, 3×4/163]

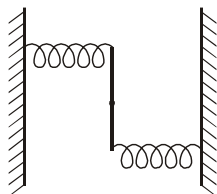
3. The net external force acting on the disk when its centre of mass is at displacement  $x$  with respect to its equilibrium position is
- (A)  $-kx$  (B)  $-2kx$  (C)  $-\frac{2kx}{3}$  (D)  $-\frac{4kx}{3}$
4. The centre of mass of the disk undergoes simple harmonic motion with angular frequency  $\omega$  equal to
- (A)  $\sqrt{\frac{k}{M}}$  (B)  $\sqrt{\frac{2k}{M}}$  (C)  $\sqrt{\frac{2k}{3M}}$  (D)  $\sqrt{\frac{4k}{3M}}$
5. The maximum value of  $V_0$  for which the disk will roll without slipping is
- (A)  $\mu g \sqrt{\frac{M}{k}}$  (B)  $\mu g \sqrt{\frac{M}{2k}}$  (C)  $\mu g \sqrt{\frac{3M}{k}}$  (D)  $\mu g \sqrt{\frac{5M}{2k}}$
6. The  $x$ - $t$  graph of a particle undergoing simple harmonic motion is shown below. The acceleration of the particle at  $t = 4/3$  s is



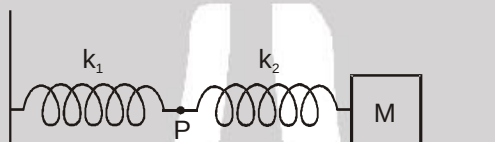
- (A)  $\frac{\sqrt{3}}{32} \pi^2 \text{ cm/s}^2$  (B)  $-\frac{\pi^2}{32} \text{ cm/s}^2$  (C)  $\frac{\pi^2}{32} \text{ cm/s}^2$  (D)  $\frac{\sqrt{3}}{32} \pi^2 - \text{cm/s}^2$



7. A uniform rod of length  $L$  and mass  $M$  is pivoted at the centre. Its two ends are attached to two springs of equal spring constants  $k$ . The springs are fixed to rigid supports as shown in the figure, and the rod is free to oscillate in the horizontal plane. The rod is gently pushed through a small angle  $\theta$  in one direction and released. The frequency of oscillation is : **[IIT-JEE 2009, 3/160, -1]**



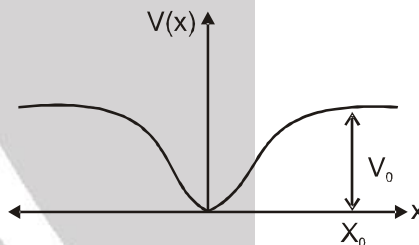
- (A)  $\frac{1}{2\pi} \sqrt{\frac{2k}{M}}$  (B)  $\frac{1}{2\pi} \sqrt{\frac{k}{M}}$  (C)  $\frac{1}{2\pi} \sqrt{\frac{6k}{M}}$  (D)  $\frac{1}{2\pi} \sqrt{\frac{24k}{M}}$
8. The mass  $M$  shown in the figure oscillates in simple harmonic motion with amplitude  $A$ . The amplitude of the point  $P$  is **[JEE 2009, 3/160, -1]**



- (A)  $\frac{k_1 A}{k_2}$  (B)  $\frac{k_2 A}{k_1}$  (C)  $\frac{k_1 A}{k_1 + k_2}$  (D)  $\frac{k_2 A}{k_1 + k_2}$

### Comprehension-2

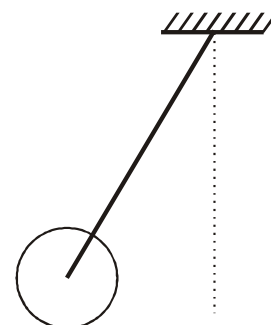
When a particle of mass  $m$  moves on the  $x$ -axis in a potential of the form  $V(x) = kx^2$ , it performs simple harmonic motion. The corresponding time period is proportional to  $\sqrt{\frac{m}{k}}$ , as can be seen easily using dimensional analysis. However, the motion of a particle can be periodic even when its potential energy increases on both sides of  $x = 0$  in a way different from  $kx^2$  and its total energy is such that the particle does not escape to infinity.



Consider a particle of mass  $m$  moving on the  $x$ -axis. Its potential energy is  $V(x) = \alpha x^4$  ( $\alpha > 0$ ) for  $|x|$  near the origin and becomes a constant equal to  $V_0$  for  $|x| \geq X_0$  (see figure) **[JEE 2010, 3×3/160, -1]**

9. If the total energy of the particle is  $E$ , it will perform periodic motion only if :  
 (A)  $E < 0$  (B)  $E > 0$  (C)  $V_0 > E > 0$  (D)  $E > V_0$
10. For periodic motion of small amplitude  $A$ , the time period  $T$  of this particle is proportional to :  
 (A)  $A \sqrt{\frac{m}{\alpha}}$  (B)  $\frac{1}{A} \sqrt{\frac{m}{\alpha}}$  (C)  $A \sqrt{\frac{\alpha}{m}}$  (D)  $\frac{1}{A} \sqrt{\frac{\alpha}{m}}$
11. The acceleration of this particle for  $|x| > X_0$  is :  
 (A) proportional to  $V_0$  (B) proportional to  $\frac{V_0}{mX_0}$   
 (C) proportional to  $\sqrt{\frac{V_0}{mX_0}}$  (D) zero



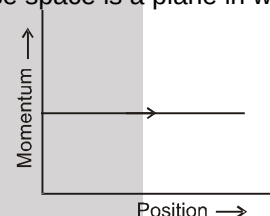


- 12\*. A metal rod of length 'L' and mass 'm' is pivoted at one end. A thin disk of mass 'M' and radius 'R' ( $R < L$ ) is attached at its center to the free end of the rod. Consider two ways the disk is attached: (case A). The disk is not free to rotate about its center and (case B) the disk is free to rotate about its center. The rod-disk system performs SHM in vertical plane after being released from the same displaced position. Which of the following statement(s) is (are) true? [JEE 2011, 4/160]

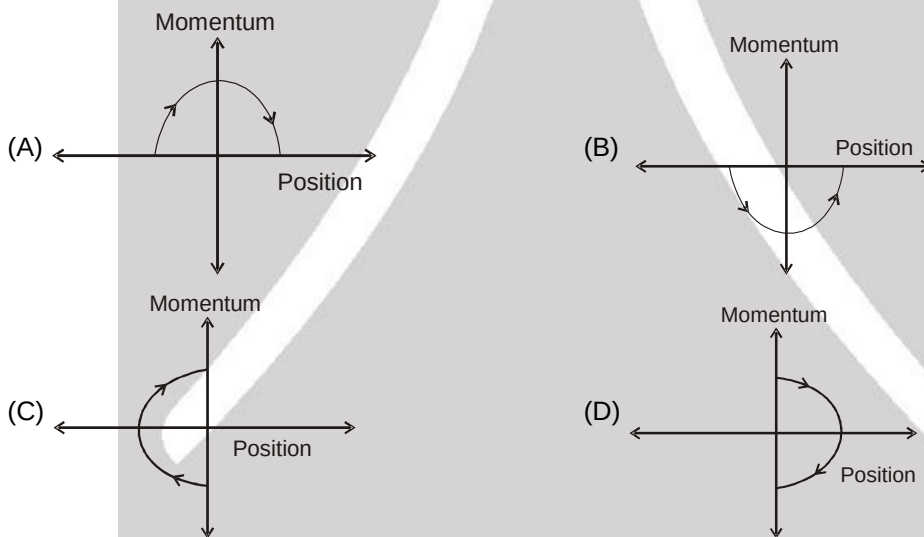
- (A) Restoring torque in case A = Restoring torque in case B  
 (B) Restoring torque in case A < Restoring torque in case B  
 (C) Angular frequency for case A > Angular frequency for case B.  
 (D) Angular frequency for case A < Angular frequency for case B.

### Comprehension-3

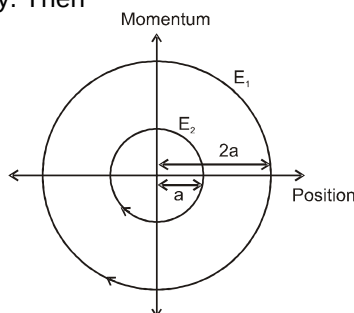
Phase space diagrams are useful tools in analyzing all kinds of dynamical problems. They are especially useful in studying the changes in motion as initial position and momentum are changed. Here we consider some simple dynamical systems in one-dimension. For such systems, phase space is a plane in which position is plotted along horizontal axis and momentum is plotted along vertical axis. The phase space diagram is  $x(t)$  vs.  $p(t)$  curve in this plane. The arrow on the curve indicates the time flow. For example, the phase space diagram for a particle moving with constant velocity is a straight line as shown in the figure. We use the sign convention in which position or momentum. Upwards (or to right) is positive and downwards (or to left) is negative. [JEE 2011, 3×3/160, -1]



13. The phase space diagram for a ball thrown vertically up from ground is :



14. The phase space diagram for simple harmonic motion is a circle centered at the origin. In the figure, the two circles represent the same oscillator but for different initial conditions, and  $E_1$  and  $E_2$  are the total mechanical energies respectively. Then

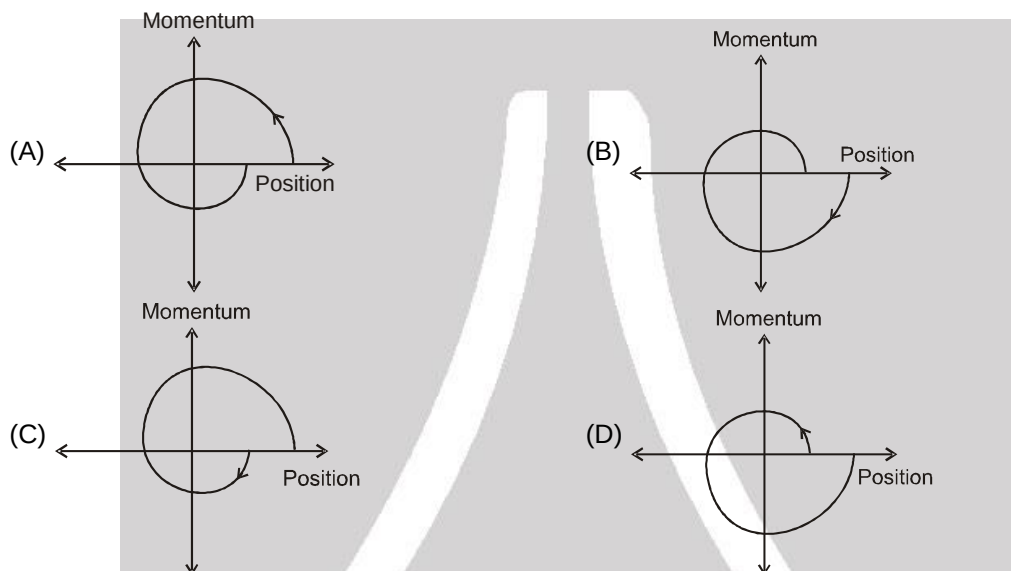
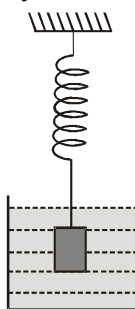


- (A)  $E_1 = \sqrt{2} E_2$  (B)  $E_1 = 2E_2$  (C)  $E_1 = 4E_2$  (D)  $E_1 = 16E_2$





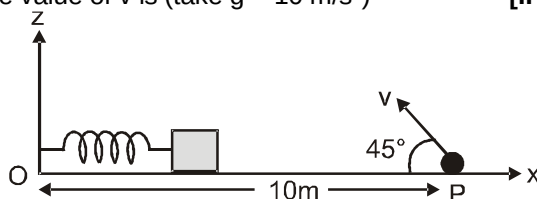
15. Consider the spring-mass system, with the mass submerged in water, as shown in the figure. The phase space diagram for one cycle of this system is :



16. A point mass is subjected to two simultaneous sinusoidal displacements in x-direction,  $x_1(t) = A \sin \omega t$  and  $x_2(t) = A \sin\left(\omega t + \frac{2\pi}{3}\right)$ . Adding a third sinusoidal displacement  $x_3(t) = B \sin(\omega t + \phi)$  brings the mass to a complete rest. The values of B and  $\phi$  are [JEE 2011, 3/160, -1]

- (A)  $\sqrt{2}A, \frac{3\pi}{4}$  (B)  $A, \frac{4\pi}{3}$  (C)  $\sqrt{3}A, \frac{5\pi}{6}$  (D)  $A, \frac{\pi}{3}$

17. A small block is connected to one end of a massless spring of un-stretched length 4.9 m. The other end of the spring (see the figure) is fixed. The system lies on a horizontal frictionless surface. The block is stretched by 0.2 m and released from rest at  $t = 0$ . It then executes simple harmonic motion with angular frequency  $\omega = \frac{\pi}{3} \text{ rad/s}$ . Simultaneously at  $t = 0$ , a small pebble is projected with speed  $v$  from point P at an angle of  $45^\circ$  as shown in the figure. Point P is at a horizontal distance of 10 cm from O. If the pebble hits the block at  $t = 1\text{s}$ , the value of  $v$  is (take  $g = 10 \text{ m/s}^2$ ) [IIT-JEE-2012, Paper-1; 3/70, -1]



- (A)  $\sqrt{50} \text{ m/s}$  (B)  $\sqrt{51} \text{ m/s}$  (C)  $\sqrt{52} \text{ m/s}$  (D)  $\sqrt{53} \text{ m/s}$



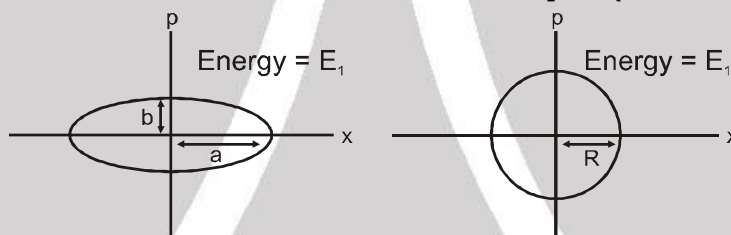
- 18\*. A particle of mass  $m$  is attached to one end of a mass-less spring of force constant  $k$ , lying on a frictionless horizontal plane. The other end of the spring is fixed. The particle starts moving horizontally from its equilibrium position at time  $t = 0$  with an initial velocity  $u_0$ . When the speed of the particle is  $0.5 u_0$ , it collides elastically with a rigid wall. After this collision :

[JEE (Advanced) 2013, 4/60]

- (A) the speed of the particle when it returns to its equilibrium position is  $u_0$  .
- (B) the time at which the particle passes through the equilibrium position for the first time is  $t = \pi \sqrt{\frac{m}{k}}$  .
- (C) the time at which the maximum compression of the spring occurs is  $t = \frac{4\pi}{3} \sqrt{\frac{m}{k}}$  .
- (D) the time at which the particle passes throughout the equilibrium position for the second time is  $t = \frac{5\pi}{3} \sqrt{\frac{m}{k}}$  .

19. Two independent harmonic oscillators of equal mass are oscillating about the origin with angular frequencies  $\omega_1$  and  $\omega_2$  and have total energies  $E_1$  and  $E_2$ , respectively. The variations of their momenta  $p$  with positions  $x$  are shown in figures. If  $\frac{a}{b} = n^2$  and  $\frac{a}{R} = n$ , then the correct equation(s) is (are) :

[JEE (Advanced) 2015 ; P-1, 4/88, -2]



- (A)  $E_1 \omega_1 = E_2 \omega_2$       (B)  $\frac{\omega_2}{\omega_1} = n^2$       (C)  $\omega_1 \omega_2 = n^2$       (D)  $\frac{E_1}{\omega_1} = \frac{E_2}{\omega_2}$

20. A particle of unit mass is moving along the  $x$ -axis under the influence of a force and its total energy is conserved. Four possible forms of the potential energy of the particle are given in column I (a and  $U_0$  are constants). Match the potential energies in column I to the corresponding statement(s) in column II.

[JEE (Advanced) 2015 ; 8/88, -1]

Column-I

- (A)  $U_1(x) = \frac{U_0}{2} \left[ 1 - \left( \frac{x}{a} \right)^2 \right]^2$
- (B)  $U_2(x) = \frac{U_0}{2} \left( \frac{x}{a} \right)^2$
- (C)  $U_3(x) = \frac{U_0}{2} \left( \frac{x}{a} \right)^2 \exp \left[ - \left( \frac{x}{a} \right)^2 \right]$
- (D)  $U_4(x) = \frac{U_0}{2} \left[ \frac{x}{a} - \frac{1}{3} \left( \frac{x}{a} \right)^3 \right]$

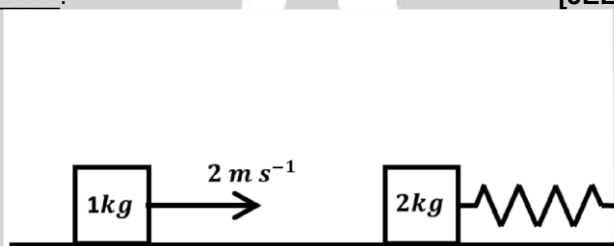
Column-II

- (P) the force acting on the particle is zero at  $x = a$ .
- (Q) the force acting on the particle is zero at  $x = 0$ .
- (R) the force acting on the particle is zero at  $x = -a$ .
- (S) The particle experiences an attractive force towards  $x = 0$  in the region  $|x| < a$
- (T) The particle with total energy  $\frac{U_0}{4}$  can oscillate about the point  $x = -a$ .



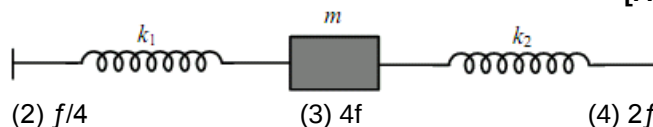


- 21\*. A block with mass  $M$  is connected by a massless spring with stiffness constant  $k$  to a rigid wall and moves without friction on a horizontal surface. The block oscillates with small amplitude  $A$  about an equilibrium position  $x_0$ . Consider two cases : (i) when the block is at  $x_0$  ; and (ii) when the block is at  $x = x_0 + A$ . In both the cases, a particle with mass  $m$  ( $< M$ ) is softly placed on the block after which they stick to each other. Which of the following statement(s) is(are) true about the motion after the mass  $m$  is placed on the mass  $M$  ? [JEE (Advanced) 2016 ; P-2, 4/62, -2]
- (A) The amplitude of oscillation in the first case changes by a factor of  $\sqrt{\frac{M}{m+M}}$ , whereas in the second case it remains unchanged  
 (B) The final time period of oscillation in both the cases is same  
 (C) The total energy decreases in both the cases  
 (D) The instantaneous speed at  $x_0$  of the combined masses decreases in both the cases
22. A spring-block system is resting on a frictionless floor as shown in the figure. The spring constant is  $2.0 \text{ Nm}^{-1}$  and the mass of the block is  $2.0 \text{ kg}$ . Ignore the mass of the spring. Initially the spring is in an unstretched condition. Another block of mass  $1.0 \text{ kg}$  moving with a speed of  $2.0 \text{ ms}^{-1}$  collides elastically with the first block. The collision is such that the  $2.0 \text{ kg}$  block does not hit the wall. The distance, in metres, between the two blocks when the spring returns to its unstretched position for the first time after the collision is \_\_\_\_\_. [JEE (Advanced) 2018, P-1, 3/60]



## PART - II : JEE (MAIN) / AIEEE PROBLEMS (PREVIOUS YEARS)

1. The maximum velocity of a particle, executing simple harmonic motion with an amplitude  $7 \text{ mm}$ , is  $4.4 \text{ m/s}$ . The period of oscillation is : [AIEEE 2006 ; 3/165, -1]  
 (1)  $100 \text{ s}$  (2)  $0.01 \text{ s}$  (3)  $10 \text{ s}$  (4)  $0.1 \text{ s}$
- 2\*. A coin is placed on a horizontal platform which undergoes vertical simple harmonic motion of angular frequency  $\omega$ . The amplitude of oscillation is gradually increased. The coin will leave contact with the platform for the first time : [AIEEE 2006 ; 3/165, -1]  
 (1) at the highest position of the platform (2) at the mean position of the platform  
 (3) for an amplitude of  $\frac{g}{\omega^2}$  (4) for an amplitude of  $\frac{g^2}{\omega^2}$
3. The displacement of an object attached to a spring and executing simple harmonic motion is given by  $x = 2 \times 10^{-2} \cos \pi t$  metres. The time at which the maximum speed first occurs is : [AIEEE 2007 ; 3/120, -1]  
 (1)  $0.5 \text{ s}$  (2)  $0.75 \text{ s}$  (3)  $0.125 \text{ s}$  (4)  $0.25 \text{ s}$
4. A point mass oscillates along the  $x$ -axis according to the law  $x = x_0 \cos(\omega t - \pi/4)$ . If the acceleration of the particle is written as,  $a = A \cos(\omega t + \delta)$ , then : [AIEEE 2007 ; 3/120, -1]  
 (1)  $A = x_0, \delta = -\pi/4$  (2)  $A = x_0 \omega^2, \delta = -\pi/4$  (3)  $A = x_0 \omega^2, \delta = -\pi/4$  (4)  $A = x_0 \omega^2, \delta = 3\pi/4$
5. Two springs, of force constants  $k_1$  and  $k_2$ , are connected to a mass  $m$  as shown. The frequency of oscillation of mass is  $f$ . If both  $k_1$  and  $k_2$  are made four times their original values, the frequency of oscillation becomes: [AIEEE 2007 ; 3/120, -1]



- (1)  $f/2$  (2)  $f/4$  (3)  $4f$  (4)  $2f$



6. A particle of mass  $m$  executes simple harmonic motion with amplitude  $a$  and frequency  $\nu$ . The average kinetic energy during its motion from the position of equilibrium to the end is : [AIEEE 2007; 3/120, -1]  
 (1)  $\pi^2 ma^2 \nu^2$  (2)  $\frac{1}{4} ma^2 \nu^2$  (3)  $4\pi^2 ma^2 \nu^2$  (4)  $2\pi^2 ma^2 \nu^2$
- 7.\* If  $x$ ,  $v$  and  $a$  denote the displacement, the velocity and the acceleration of a particle executing simple harmonic motion of time period  $T$ , then, which of the following does not change with time? [AIEEE 2009; 3/120, -1]  
 (1)  $\frac{aT}{x}$  (2)  $aT + 2\pi v$  (3)  $\frac{aT}{v}$  (4)  $a^2 T^2 + 4\pi^2 v^2$
8. A mass  $M$ , attached to a horizontal spring, executes SHM with a amplitude  $A_1$ . When the mass  $M$  passes through its mean position then a smaller mass  $m$  is placed over it and both of them move together with amplitude  $A_2$ . The ratio of  $\left(\frac{A_1}{A_2}\right)$  is : [AIEEE - 2011, 4/120, -1]  
 (1)  $\frac{M}{M+m}$  (2)  $\frac{M+m}{M}$  (3)  $\left(\frac{M}{M+m}\right)^{1/2}$  (4)  $\left(\frac{M+m}{M}\right)^{1/2}$
9. If a spring of stiffness ' $k$ ' is cut into two parts 'A' and 'B' of length  $\ell_A : \ell_B = 2 : 3$ , then the stiffness of spring 'A' is given by : [AIEEE 2011, 11 May; 4/120, -1]  
 (1)  $\frac{3k}{5}$  (2)  $\frac{2k}{5}$  (3)  $k$  (4)  $\frac{5k}{2}$
10. Two particles are executing simple harmonic motion of the same amplitude  $A$  and frequency  $\omega$  along the  $x$ -axis. Their mean position is separated by distance  $X_0 (X_0 > A)$ . If the maximum separation between them is  $(X_0 + A)$ , the phase difference between their motion is : [AIEEE - 2011, 4/120, -1]  
 (1)  $\pi/2$  (2)  $\pi/3$  (3)  $\pi/4$  (4)  $\pi/6$
11. If a simple pendulum has significant amplitude (up to a factor of  $1/e$  of original) only in the period between  $t = 0$  s to  $t = \tau$  s, then  $\tau$  may be called the average life of the pendulum. When the spherical bob of the pendulum suffers a retardation (due to viscous drag) proportional to its velocity, with ' $b$ ' as the constant of proportionality, the average life time of the pendulum is (assuming damping is small) in seconds : [AIEEE 2012; 4/120, -1]  
 (1)  $\frac{0.693}{b}$  (2)  $b$  (3)  $\frac{1}{b}$  (4)  $\frac{2}{b}$
12. The amplitude of a damped oscillator decreases to 0.9 times its original magnitude in 5s. In another 10s it will decrease to  $\alpha$  times its original magnitude, where  $\alpha$  equals. [JEE (Main) 2013, 4/120]  
 (1) 0.7 (2) 0.81 (3) 0.729 (4) 0.6
13. A particle moves with simple harmonic motion in a straight line. In first  $\tau$  s, after starting from rest it travels a distance  $a$ , and in next  $\tau$  s it travels  $2a$ , in same direction, then : [JEE (Main) 2014; 4/120, -1]  
 (1) amplitude of motion is  $3a$  (2) time period of oscillations is  $8\tau$   
 (3) amplitude of motion is  $4a$  (4) time period of oscillations is  $6\tau$
14. For a simple pendulum, a graph is plotted between its kinetic energy (KE) and potential energy (PE) against its displacement  $d$ . Which one of the following represents these correctly? (graphs are schematic and not drawn to scale) [JEE (Main) 2015; 4/120, -1]
- (1)

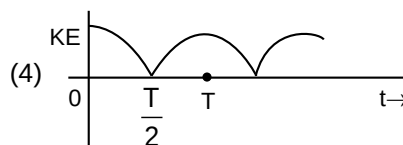
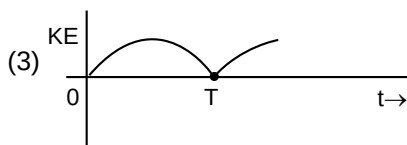
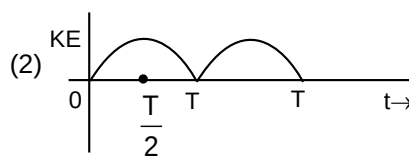
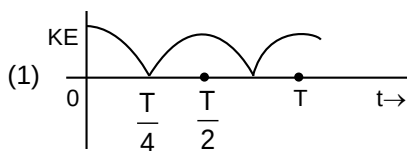
(2)

(3)

(4)
15. A particle performs simple harmonic motion with amplitude  $A$ . Its speed is trippled at the instant that it is at distance  $2A/3$  from equilibrium position. The new amplitude of the motion is. [JEE (Main) 2016; 4/120, -1]  
 (1)  $3A$  (2)  $\sqrt{3} A$  (3)  $\frac{7A}{3}$  (4)  $\frac{A}{3} \sqrt{41}$



16. A particle is executing simple harmonic motion with a time period  $T$ . At time  $t = 0$ , it is at its position of equilibrium. The kinetic energy-time graph of the particle will look like : [JEE (Main) 2017; 4/120, -1]



17. A silver atom in a solid oscillates in simple harmonic motion in some direction with a frequency of  $10^{12}$ /sec. What is the force constant of the bonds connecting one atom with the other ? (Mole wt. of silver = 108 and Avagadro number =  $6.02 \times 10^{23}$  gm mole $^{-1}$ ) [JEE (Main) 2018; 4/120, -1]
- (1) 2.2 N/m (2) 5.5 N/m (3) 6.4 N/m (4) 7.1 N/m

## Answers

### EXERCISE-1

#### PART - I

#### Section (A) :

- A-1. Amplitude = 5 m, Phase constant =  $\frac{\pi}{6}$ , Time period = 2 s, Maximum speed =  $5\pi$  m/s
- A-2. (a) 2.0 cm,  $\frac{\pi}{50}$  s = 0.063 s, 100 N/m  
(b) 1.0 cm,  $\sqrt{3}$  m/s, -100 m/s $^2$
- A-3. (a) T/12, (b) T/8, (c) T/6, (d) T/4, (e) T/8
- A-4.  $\frac{\sqrt{3}v_0}{2}$
- A-5.  $x = (10\text{cm}) \sin \left[ \left( \frac{\pi}{3} \text{s}^{-1} \right) t + \frac{\pi}{6} \right]$ ,  $\frac{10}{9} \pi^2 \approx 11 \text{ cm/s}^2$
- A-6.  $\pm \frac{6}{5} \text{ cm} = \pm 1.2 \text{ cm}$  from the mean position

#### Section (B) :

- B-1.  $\pm 5 \text{ cm}$  B-2.  $A = 0.06 \text{ m}$

#### Section (C) :

- C-1. 0.1 N/m C-2. 40 J
- C-3.  $\frac{16}{10\pi^2} = 0.16 \text{ Kg}$
- C-4. (a)  $\frac{F}{k}$ ,  $2\pi\sqrt{\frac{M}{k}}$ , (b)  $\frac{F^2}{2k}$  (c)  $\frac{F^2}{2k}$

- C-5. (a)  $2\pi\sqrt{\frac{m}{k_1+k_2}}$ ,  $k_{\text{eq.}} = k_1 + k_2$ ; (b)

$$2\pi\sqrt{\frac{m}{k_1+k_2}}, k_{\text{eq.}} = k_1 + k_2; \text{ (c)}$$

$$2\pi\sqrt{\frac{m(k_1+k_2)}{k_1k_2}}, k_{\text{eq.}} = \frac{k_1k_2}{k_1+k_2}$$

Answers will remain same

- C-6.  $\frac{M^2g^2}{2k}$ ,  $\frac{M^2g^2}{4k}$  and  $\frac{M^2g^2}{6k}$  from above,  
time period =  $2\pi\sqrt{\frac{11M}{6k}}$

#### Section (D) :

- D-1. 1 m D-2. 0.25 m
- D-3.  $\left( \frac{3600}{3601} \right)^2 g = 9.794 \text{ m/s}^2$
- D-4. (i)  $2T_0$  (ii) 3g upwards

#### Section (E) :

- E-1. (a)  $T = 2\pi\sqrt{\frac{7\ell}{12g}}$  (b)  $2\pi\sqrt{\frac{2r}{g}}$   
(c)  $2\pi\sqrt{\frac{\sqrt{8}a}{3g}}$  (d)  $2\pi\sqrt{\frac{3r}{2g}}$
- E-2. (a)  $2\pi\sqrt{\frac{r\sqrt{2}}{g}}$ ,  $r/\sqrt{2}$  (b)  $2\pi\sqrt{\frac{L}{3g}}$ ,  $\frac{L}{2\sqrt{3}}$

#### Section (F) :

- F-1. (a) 7 cm (b)  $\sqrt{37} \text{ cm} = 6.1 \text{ cm}$  (c) 5 cm
- F-2.  $a\sqrt{4+2\sqrt{3}}$  F-3.  $2x^2 + \frac{y}{2} = 1$

**Section (G) :**

- G-1. Both amplitude and energy of the particle can be maximum only in the case of resonance.  
For resonance to occur,  $\omega_1 = \omega_2$
- G-2. (a) 0.3 s (b) 6.93 s (c) 3.4 s

**PART - II****Section (A) :**

- A-1. (A) A-2. (A) A-3. (B)  
A-4. (A) A-5. (A) A-6. (C)  
A-7. (A) A-8. (B) A-9. (B)  
A-10. (B)

**Section (B) :**

- B-1. (B) B-2. (C) B-3. (B)  
B-4. (C) B-5. (A) B-6. (C)

**Section (C) :**

- C-1. (D) C-2. (C) C-3. (C)  
C-4. (A) C-5. (D) C-6. (D)  
C-7. (B) C-8. (D)

**Section (D) :**

- D-1. (A) D-2. (D) D-3. (C)  
D-4. (D) D-5. (C)

**Section (E) :**

- E-1. (D) E-2. (B)

**Section (F) :**

- F-1. (B) F-2. (B) F-3. (D)  
F-4. (B)

**Section (G) :**

- G-1. (D) G-2. (C)

**PART - III**

1. (A)  $\rightarrow$  r, (B)  $\rightarrow$  p, (C)  $\rightarrow$  q, (D)  $\rightarrow$  s  
2. (A)  $\rightarrow$  p ; (B)  $\rightarrow$  q ; (C)  $\rightarrow$  p ; (D)  $\rightarrow$  s

**EXERCISE-2****PART - I**

1. (A) 2. (C) 3. (B)  
4. (C) 5. (C) 6. (A)  
7. (A) 8. (A) 9. (A)  
10. (A) 11. (B) 12. (C)  
13. (C) 14. (B) 15. (D)

**PART - II**

1. 10 2. 2 3. 2  
4. 15 5. 2 6. 16  
7. 3 8. 3 9. 12  
10. 17 11. 15 12. 8  
13. 15 14. 2 15. 8  
16. 75

**PART - III**

1. (AD) 2. (ABC) 3. (BC)  
4. (ABCDE) 5. (BD) 6. (ABCD)  
7. (BCD) 8. (ABC) 9. (BD)  
10. (BC) 11. (ACD) 12. (AC)

**PART - IV**

1. (A) 2. (C) 3. (D)  
4. (D) 5. (A) 6. (B)  
7. (B) 8. (C)

**EXERCISE-3****PART - I**

1. (ABD)  
2. (A)  $\rightarrow$  (p); (B)  $\rightarrow$  (q, s); (C)  $\rightarrow$  (s); (D)  $\rightarrow$  (q)  
3. (D) 4. (D) 5. (C)  
6. (D) 7. (C) 8. (D)  
9. (C) 10. (B) 11. (D)  
12. (AD) 13. (D) 14. (C)  
15. (B) 16. (B) 17. (A)  
18. (AD) 19. (BD)  
20. (A)  $\rightarrow$  P, Q, R, T ; (B)  $\rightarrow$  Q, S ; (C)  $\rightarrow$  P, Q, R, S ;  
(D)  $\rightarrow$  P, R, T  
21. (ABD) 22. 2.09

**PART - II**

1. (2) 2. (1, 3) 3. (1)  
4. (4) 5. (4) 6. (1)  
7. (1, 4) 8. (4) 9. (4)  
10. (2) 11. (4) 12. (3)  
13. (4) 14. (2) 15. (3)  
16. (1) 17. (4)

